# HF Communication System (S.O.O. 6215) – General data

## 1. General

The High Frequency (HF) transceiver system is used for long distance voice and data communication to other aircraft and/or ground stations.

## 2. Description

### A. High Frequency (HF) Communication System

The HF radio is controlled by a PS 440 control head that is mounted on the instrument panel, directly below the clock, on the right hand side of the multi-function displays. The audio functions of the HF radio are fully integrated into the KMA 29 audio control panel. In all other respects, the HF radio is entirely independent of the Apex avionics systems.

The KHF 1050 HF Communication System (also known as a Primus HF 1050) consists of three individual units: a KAC 1052 Antenna Coupler, a KPA 1052 Power Amplifier, and a KRX 1053 Receiver/Exciter. All three of these are installed in the aft avionics racks, between stations 406 and 451. Receiver and low level transmitter signals are located in the Receiver/Exciter. The Receiver/Exciter employs a master oscillator, which requires a short warm-up period when operating at cool temperatures.

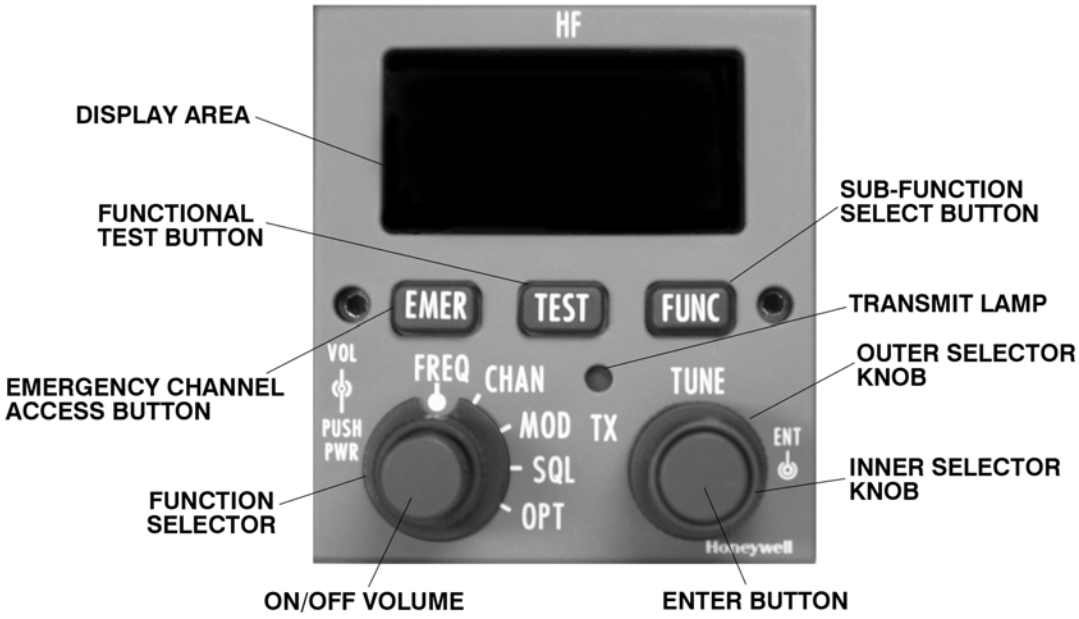
The Power Amplifier amplifies the excitation signal from the Receiver/Exciter to 200 watts peak effective power, or 50 watts of carrier power. The amplified signal is then routed to the Antenna Coupler, which matches the various impedances of the inverted V wire antenna to the 50 ohm output of the transmitter. When in receive mode, the signals from the antenna pass through the Antenna Coupler to the Receiver/Exciter via the Power Amplifier.

The inverted V antenna configuration receives and transmits signals in the frequency band from 2 MHz to 29.9999 MHz. The antenna is connected at the antenna coupler, then to an isolator at the top of the vertical stabilizer. It then connects to an antenna mast mounted on top of the aircraft.

28 Volt DC power is protected by a 5 amp RX/EX ANT CPLR (2S) (Hot Battery Bus), a 30 amp POWER AMP (2R) (Hot Battery Bus), and a 2 amp HF CTRL (4P) (Right Bus), circuit breakers located either on the avionics circuit breaker panel or on the rear circuit breaker panel located aft side of station 332.00 on the co-pilot's side (refer to 24-00-00, Electrical Power – General Data).

The HF (KHF 1050) transceiver interfaces to the following aircraft systems:

1) Dual KMA 29 Audio Control Panel. (Refer to 23-53-00, Audio Control Panel – General Data).




HF Controller – Layout
Figure 1

3. Adjustment/Test

A. Function Test HF (KHF 2100) Transceiver

1) General

These procedures are for the function test of the HF (KHF 2100) transceiver.

**CAUTION** **SOME ELECTRICAL POWER UNITS (EPU) MAY REGULATE POORLY WHEN SUBJECTED TO STRONG RF FIELDS. IF FLUCTUATIONS IN THE EPU ARE NOTED WHEN TRANSMITTING, DISCONTINUE USING THE EPU. WHEN AN EPU CANNOT BE USED, AN ALTERNATIVE METHOD OF TESTING THE HF IS WITH ENGINES RUNNING AND GENERATORS ON.**

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that the RX/EX ANT CPLR (2S), POWER AMP (2R) and the HF CTRL circuit breakers are closed.

d) On the HF controller, press the TEST button and ensure no fault codes are displayed.

e) On the HF controller, push the PWR inner concentric knob.

f) Tune to a known HF communications station. Ensure audio can be heard clearly. Adjust the volume to an acceptable level.

g) Select to a channel in which the aircraft is authorized to transmit.

h) Momentarily key the HF transceiver. Ensure the TX indicator stops flashing when the tuning process is complete.

i) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – HF Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to empennage through rear baggage compartment.

3) Locate HF Antenna aft of station 451.00 on the ceiling of the aircraft.

4) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the HF Feed Thru Assembly metal casing with the second probe on the HF ground strap, GRD 134. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

5) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

6) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structues – General data.

C. HF Coupler (KAC 1052) Pressurization Test

1) General

The KAC 1052 is pressurized with nitrogen. On an annual basis, the pressure of the KAC 1052 must be measured by using a voltmeter, a precision gauge or by the KHF 1052 Maintenance Program (if available).

2) Pressurization Test

a) Pressurization Test Procedure – Using Voltmeter

Note Pressure reading can differ by up to 0.375 volts (3 psi) due to the effects of temperature and barometric pressure at the time the readings are taken. Average multiple readings result in higher accuracy.

Note In the event that the measured voltage between Pin 7(+) and Pin 4(–) drops to 2.5 Volt DC or less between annual maintenance check, (pressure leaked to 4 psi or less), more frequent monitoring of the pressure is advised.

Note If rapid pressure loss persists, remove the HF Coupler (KAC 1052) unit from the plane and check for leaks.

1 Remove the protective cover from TEST J10522 port in front of the HF Coupler (KAC 1052).

2 Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3 Connect the voltmeter, the positive (+) lead to P10522 Pin 7 and the negative lead (–) to P10522 Pin 4. A voltmeter reading of 3.0 to 3.3 Volt DC indicates that the HF Coupler (KAC 1052) pressure is acceptable (between 8 to 10 psi).

4 If the pressure reads below 8 psi, charge the KAC 1052 unit with nitrogen through the Pressure Input and Measurement Valve to bring the pressure up to between 8 to 10 psi.

5 Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

6 Replace the protective cover on TEST J10522 port in front of the HF Coupler (KAC 1052).

7 Ensure protective cover is secure.

8 Visually inspect HF Coupler (KAC 1052) connectors to ensure they are sound and secure.

b) Pressurization Test Procedure – Using Precision Gauge

Note Pressure readings can differ by up to 3 psi due to the effects of temperature and barometric pressure at the time the readings are taken. Average multiple readings result in higher accuracy.

Note In the event that the measured pressure is 4 psi or less between annual maintenance check, more frequent monitoring of the pressure is advised.

Note If rapid pressure loss persists, remove the HF Coupler (KAC 1052) unit from the plane and check for leaks.

1 Connect the precision gauge on the Pressure Input and Measurement valve adjacent to J10522. A reading of 8 to 10 psi on the gauge indicates that the HF Coupler (KAC 1052) pressure is acceptable.

2 If the pressure reads below 8 psi, charge the KAC 1052 unit with nitrogen through the Pressure Input and Measurement valve to bring the pressure up to between 8 to 10 psi.

3 Remove the precision gauge.

c) Pressurization Test Procedure – Using KHF 1052 Maintenance Program

Note The KHF 1052 Maintenance Program is an additional software diagnostic program for the HF 1050 HF System, and is made available through Honeywell. Refer to Honeywell System Installation Manual – KHF 1050 HF Communication System, Manual No. 006-10640-0006, Rev 6 Nov/2007 or later for more details.

Note Pressure readings can differ by up to 3 psi due to the effects of temperature and barometric pressure at the time the readings are taken. Average multiple readings result in higher accuracy.

Note In the event that the measured pressure is 4 psi or less between annual maintenance check, more frequent monitoring of the pressure is advised.

Note If rapid pressure loss persists, remove the HF Coupler (KAC 1052) unit from the plane and check for leaks.

1 Remove the protective cover from TEST J10522 port in front of the HF Coupler (KAC 1052).

2 Connect the KAC 1052 to the Ground Equipment Laptop as instructed Honeywell System Installation Manual – KHF 1050 HF Communication System, Manual No. 006-10640-0006, Rev 6 Nov/2007 or later.

3 Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4 Access the ACP Maintenance page within the KHF 1050 Maintenance Program. Internal pressure is displayed in the upper right portion of the screen.

5 Confirm that the HF Coupler Pressure (KAC 1052) reading is between 8 to 10 psi.

6 To obtain multiple readings, click on Gauge Pressure, and repeat at desired intervals.

7 If the pressure reads below 8 psi, charge the KAC 1052 unit with nitrogen through the Pressure Input and Measurement valve to bring the pressure up to between 8 to 10 psi.

8 Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

9 Disconnect the KAC 1052 from the ground Equipment Laptop.

10 Replace the protective cover on TEST J10522 port in front of the HF Coupler (KAC 1052).

11 Ensure cover is secure.

12 Verify all cable connections from the HF Coupler (KAC 1052) are sound and secure.

4. Removal/Installation

**WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

**CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LINE REPLACEABLE UNIT (LRU) MOUNTING TRAY OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.**

**SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W. BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.**

A. Removal and Installation of the HF Controller (PS–440)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the HF CTRL (4P) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Release the two fasteners on the controller and remove the controller from the panel.

d) Disconnect the electrical connector from the controller.

Note Stow connectors as required.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the HF CTRL (4P) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clip.

c) Connect the electrical connector to the controller.

d) Slide the controller into the instrument panel.

e) Tighten the two fasteners on the controller.

f) Remove safety clip and close HF CTRL (4P) circuit breaker on the avionics circuit breaker panel.

g) Test the HF transceiver as per Para 3.A.

B. Removal and Installation of the Receiver/Exciter (KRX 1053)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the HF RX/EX ANT CPLR (2S) circuit breaker on the rear circuit breaker panel and fit safety clip.

c) Gain access to the rear avionics shelves through the rear baggage compartment.

d) Disconnect the electrical connector from the receiver/exciter.

e) Loosen retaining screw clamp from the front and slide the unit out of the mounting tray.

f) Remove unit from aircraft.

Note Stow connectors as required.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the HF RX/EX ANT CPLR (2S) circuit breaker on the rear circuit breaker panel is opened and fitted with safety clip.

c) Slide the unit into the mounting tray. Tighten the retaining screw clamp at the front lip of the receiver/exciter unit.

d) Reconnect the electrical connector to the receiver/exciter unit. Ensure connections are secure.

e) Remove safety clip and close HF RX/EX ANT CPLR (2S) circuit breaker on rear circuit breaker panel.

f) Test the HF transceiver as per Para 3.A.

C. Removal and Installation of the HF Antenna Coupler (KAC 1052)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the HF RX/EX ANT CPLR (2S) circuit breaker on the rear circuit breaker panel and fit safety clip.

c) Gain access to the rear avionics shelves through the rear baggage compartment.

d) Remove the nut and washer and disconnect the bonding cable from the antenna coupler.

e) Disconnect the electrical connector from the antenna coupler.

f) Loosen the two retaining screw clamps from the front and slide the unit out of the mounting tray.

g) Remove unit from aircraft.

Note Stow connectors as required.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the HF RX/EX ANT CPLR (2S) circuit breaker on the rear circuit breaker panel is opened and fitted with safety clip.

c) Slide the unit into the mounting tray. Tighten the two retaining screw clamps at the front lip of the antenna coupler unit.

d) Connect the bonding cable to the antenna coupler and tighten the nut and washer.

e) Reconnect the electrical connector to the antenna coupler unit. Ensure connections are secure.

f) Remove safety clip and close HF RX/EX ANT CPLR (2S) circuit breaker on the rear circuit breaker panel.

g) Test the HF transceiver as per Para 3.A.

D. Removal and Installation of the HF Power Amplifier (KPA 1052)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the POWER AMP (2R) circuit breaker on the rear circuit breaker panel and fit safety clip.

c) Gain access to the rear avionics shelves through the rear baggage compartment.

d) Disconnect the electrical connector from the power amplifier.

e) Loosen retaining screw clamp from the front and slide the unit out of the mounting tray.

f) Remove unit from aircraft.

Note Stow connectors as required.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the POWER AMP (2R) circuit breaker on the rear circuit breaker panel is opened and fitted with safety clip.

c) Slide the unit into the mounting tray. Tighten the retaining screw clamp at the front lip of the power amplifier unit.

d) Reconnect the electrical connector to the power amplifier unit. Ensure connections are secure.

e) Remove safety clip and close POWER AMP (2R) circuit breaker on the rear circuit breaker panel.

f) Test the HF transceiver as per Para 3.A.

E. Removal and Installation of the HF Antenna Assembly (Mod 6/2368)

1) Removal

Note Removal of the wire antenna can only be accomplished by cutting and discarding the wire.

a) Due to shock hazard from the HF transmitter, ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the RX/EX ANT ANT CPLR (2S), POWER AMP (2R), HF CTRL (4P) circuit breakers and fit safety clips.

c) Cut the wire, not less than 2 inches from the fitting and remove the brown polyethylene, exposing the wire gripped by the jaws.

d) Slide the extractor (Dayton Granger tool part number 14053 of 50ARM300–250) onto the bared wire aircraft and push the extractor against the jaws to release the antenna cable. Repeat the procedure at the opposite end of the wire. Discard the wire.

e) Disconnect the co-axial connector from the underside of the antenna feedthrough.

Note Stow connectors as required.

f) Remove the antenna cable feed-thru assemblies screws (6) and washers (6).

g) Remove the mounting assembly from the vertical stabilizer bracket.

2) Installation

a) Due to shock hazard from the HF transmitter, ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the RX/EX ANT ANT CPLR (2S), POWER AMP (2R), HF CTRL (4P) circuit breakers are opened and fitted with safety clips.

c) Install the vertical stabilizer bracket mounting assembly with the antenna wire fed through.

d) Mount the antenna cable feed-thru and tighten the screws (6).

e) Carefully inspect the antenna wire for damaged insulation. Straighten the wire carefully to eliminate all kinks.

Note Handle the wire carefully to keep it free of metallic particles or similar material that might become imbedded in the wire and thereby reduce dielectric strength.

f) Cut the wire to approximate length, allowing a few extra inches for later trimming to exact length.

g) Use Dayton Granger tool part number 14053 or 50ARM300–250. Insert the insulated wire into the tool until it bottoms out.

h) Hold wire in one hand. With other hand place thin knife blade in slot of the tool. Apply pressure on the knife blade and rotate a complete revolution around the wire to partially sever the insulation 7/17 inches from the end of the wire. The tool (part number 14053 or 50ARM300–250) is designed to prevent the knife blade from making contact with the conductor. This is important, since radial nicking of the conductor may cause early failure of the wire.

i) Hold wire in one hand and with pliers grip and twist end of insulation, removing the end of insulation. The exposed conductor must be 7/16", with no nicks or scratches.

j) Using a file, carefully remove all burrs and sharp edges from the end of the conductor to permit easy entry into the chuck jaws.

k) At fuselage feedthrough, thread the wire through the sleeve and elbow.

l) Thrust the wire end firmly and quickly into the feedthrough fitting, through the hole as far as it will go. This will permit the chuck to engage the bared end of the wire. Make sure that the entire 7/16" length of bare conductor is firmly secured by jaws.

m) To ensure firm seating of wire, give sharp tug outward from the jaws.

n) Fill the threaded cavity with Dow Corning DC–4 silicone grease. Screw elbow into insulator body, forcing the DC–4 into the elbow.

o) Fill the elbow with DC–4, and screw the sleeve into the elbow forcing the grease into the sleeve, such that the grease flows out the top opening of the sleeve. Sleeve should be screwed firmly into elbow, finger tight.

Note In order to prevent the entry of moisture, the antenna feedthrough sleeve must be packed with Dow Corning DC–4 Silicone Grease. Cleanliness, adequate quantity of “packed” DC–4 Grease, and a tight connection are of critical importance.

Note To improve finger pressure on sleeve, a small piece of emery cloth between fingers and sleeve is permitted. Do NOT use pliers.

p) Thread the wire through the tensioner on the vertical stabilizer and forward to the mast of the forward fuselage.

q) Trim wire to length. After installation with tensioners under load, the tension should be 35 ± 5 lbs.

r) Strip 7/16 inch of insulation from the wire per Para 4.E.2).h) and Para 4.E.2).i).

s) Thrust the wire end firmly and quickly into the shackle through the hole as far as it will go. This will permit the chuck to engage the bared end of the wire.

t) To ensure firm seating of wire, give sharp tug outward. When wire is properly installed, the end of insulation should pass into counter bore of chuck housing.

u) Attach the shackle to the mast.

v) Verify tension is 35 ± 5 lbs.

w) Remove safety clips and close RX/EX ANT CPLR (2S), POWER AMP (2R), HF CTRL (4P) circuit breakers.

x) Test the HF transceiver as per Para 3.A.

5. Troubleshooting

A. HF (KHF 2100) Transceiver

1) If the HF (KHF 2100) transceiver is deemed inoperative check the following:

a) Ensure RX/EX ANT CPLR (2S), POWER AMP (2R), HF CTRL (4P) circuit breakers are closed.

b) Check that the system units are securely installed and the retaining mechanism is securely tightened.

c) Visually inspect receiver/exciter, antenna coupler, power amplifier, and controller connectors for damaged wires or contacts.

d) If troubleshooting at the wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

e) If repair or service is required, send to appropriate rated avionics repair facility.

Note The HF (KHF 1050) system components contain BITE (Built-In Test Equipment) which constantly monitors the operation health of the unit. The KHF 1050 also stores detected failures in non volatile memory for later review.

Note If appropriately rated for avionics repair, fault isolation procedures are contained in the KHF 1050 CMM.

B. HF Controller (PS440 Control Head)

| PS440 Displays | Possible Cause | Check | Remedial Action |
|---|---|---|---|
| PRS W<br>(Antenna Coupler Pressure Warning)<br><br>Note The HF System will continue to operate normally. | – the pressure has dropped significantly below normal<br>– may indicate that the antenna coupler is approaching a Pressure Fault Condition. | – the pressure of the Antenna Coupler | – charge the Antenna Coupler with nitrogen to 8–10 psi through the pressure input and measure valve |
| PRS F<br>(Antenna Coupler Pressure Fault)<br><br>Note The HF System will operate at low-power condition, (50W), regardless of the transmit power selected by the pilot. | – the pressure has dropped to a level where there is a risk of arcing inside the antenna coupler | – the pressure of the Antenna Coupler | – charge Antenna Coupler with nitrogen to 8–10 psi through the pressure input and measure valve |
| PANEL<br>(Control Panel Failure) | – if HF Controller (PS440 Control Head) panels has an internal failure | | – replace with serviceable unit |
| FAIL RXEX<br>(Receiver/Exciter Failure) | – there is a failure of the Receiver/Exciter Unit<br>– failure in the connections to the unit<br>– loss of DC power to unit<br>– communications problems between the Receiver/Exciter and the Controller | – the connectors and pins are in good condition and are fully mated with the units<br>– that DC Power is available at the KRX 1053 connector<br>– continuity in the wiring between the KRX 1053 and PS440 | – replace with the serviceable unit |

| PS440 Displays | Possible Cause | Check | Remedial Action |
|---|---|---|---|
| FAIL PA<br>(Power Amplifier Failure) | – failure of the Power Amplifier unit<br>– failure in the connections to unit<br>– loss of DC power to unit | – the connectors and pins are in good and condition and are fully mated with the units<br>– that DC Power is available at the KPA 1052 connector | – replace with serviceable unit |
| FAIL CPLR<br>(Antenna Coupler Failure) | – failure of the Antenna Coupler unit<br>– failure in connections to unit<br>– loss of DC power to unit<br>– communications problems between the Antenna Coupler and the Receiver/Exciter | – the connectors and pins are in good and condition and are fully mated with the units<br>– that DC Power is available at the KRX 1053 connector<br>– for continuity in the wiring between the KAC 1052 and KRX 1053 | – replace with serviceable unit |

1) If repair or service is required, send to appropriate rated avionics repair facility.

Note The HF (KHF 1050) system components contain BITE (Built-In Test Equipment) which constantly monitors the operation health of the unit. The KHF 1050 also stores detected failures in non volatile memory for later review.

Note If appropriately rated for avionics repair, shop fault isolation procedures are contained in KHF 1050 CMM.

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